Foundations & Reasoning

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

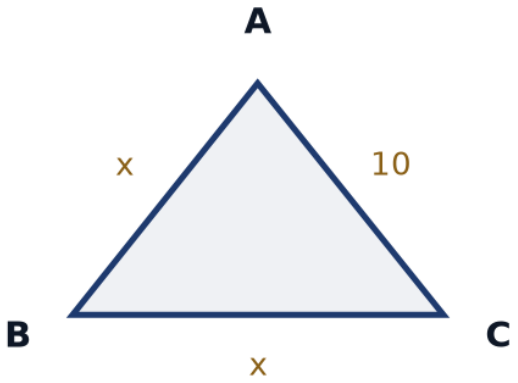
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Points, lines, and planes

Two angles, $\angle A$ and $\angle B$, are complementary. If $\angle A = (3x + 15)^\circ$ and $\angle B = (2x + 30)^\circ$, what is the measure of $\angle A$?

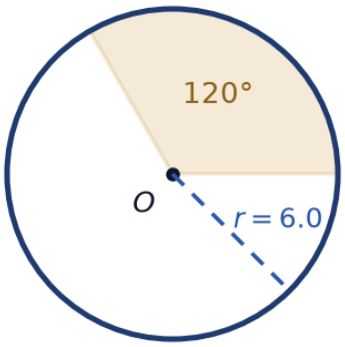
2 [Foundational] • Target: Line segments and distance

In the isosceles triangle ABC shown below, AB = AC. If the measure of angle BAC is 52°, what is the measure of angle ABC?



3 [Foundational] • Target: Points and midpoints

In the given circle, the central angle $\angle AOB$ measures 120°. What is the measure of the inscribed angle $\angle ACB$ that subtends the same arc AB?



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4 [Foundational] • Target: Review: Slopes of lines

Consider the line L with the equation $3x - 4y + 15 = 0$. A point P has coordinates $(5, -2)$. Calculate the shortest perpendicular distance from point P to line L.

5 [Foundational] • Target: Review: Angles and angle measure

In ΔABC, points M and N are the midpoints of sides AB and AC respectively. If the area of ΔAMN is $18\ \text{cm}^2$, what is the area of trapezoid $BMNC$?

6 [Foundational] • Target: Review: Angle relationships

In the figure below, lines AB and CD are parallel with $AB = 10$ and $CD = 15$. Segments AD and BC intersect at point E. If the perimeter of ΔABE is 26, what is the perimeter of ΔCDE?

7 [Foundational] • Target: Points, lines, and planes

Chords AB and CD intersect inside a circle at point P. The segments on chord AB measure $PA = x$ and $PB = x + 7$. The segments on chord CD measure $PC = 3$ and $PD = 6$. What is the total length of chord AB?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Line segments and distance

In ΔABC, $\angle A = 50^\circ$. Transversal DE is drawn such that $\angle ADE = \angle C = 65^\circ$. What is the measure of $\angle B$?

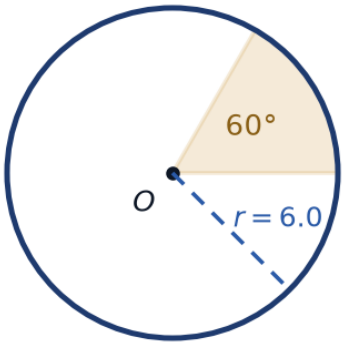
9 [Intermediate] • Target: Points and midpoints

In an hourglass figure with $AB \parallel CD$, $\angle BAE = 38^\circ$ and $\angle AEB = 84^\circ$. What is the measure of $\angle EDC$?

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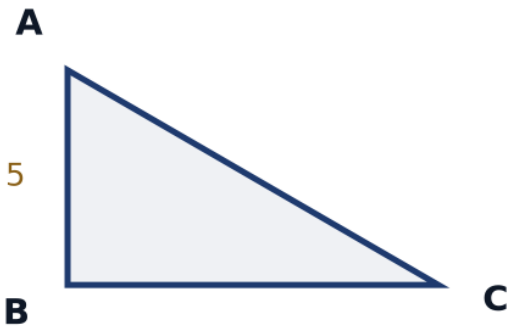
10 [Intermediate] • Target: Review: Slopes of lines

A circle has a radius of 6. A sector of the circle is formed by a central angle of 60°. What is the area of the sector?



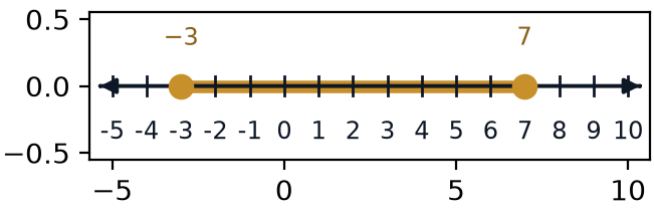
11 [Intermediate] • Target: Review: Angles and angle measure

In right triangle ABC, $\angle B = 90^\circ$, $\angle A = 60^\circ$, and the length of side AB is 5. What is the length of side BC?



12 [Intermediate] • Target: Review: Angle relationships

On a number line, point A is at -3, and point B is at 7. Point M is the midpoint of segment $\overline{AB}$. What is the coordinate of point M?



13 [Intermediate] • Target: Points, lines, and planes

In the given right triangle ABC, where angle C is the right angle, the length of side AC is 5 units and the length of side BC is 5 units. What is the exact length of the hypotenuse AB?

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14 [Intermediate] • Target: Line segments and distance

In ΔABC, line segment DE is parallel to BC, with point D lying on $\overline{AB}$ and point E lying on $\overline{AC}$. If $AD = 3$, $DB = 6$, and $AE = 5$, what is the length of segment $\overline{EC}$?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Points and midpoints • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

In the coordinate plane, the directed line segment from $A(-5, -3)$ to $B(5, 7)$ is partitioned by point P in the ratio $3:2$. What are the coordinates of point P?

16 [Challenging] • Target: Review: Slopes of lines • Cross-Lesson: 5.01 Right Triangles & Trigonometry

A student whose eyes are 1.5 m above the ground places a mirror flat on the ground 2 m in front of her feet. Looking into the mirror, she sees the reflection of the top of a school flagpole located 18 m beyond the mirror. If the angle of incidence equals the angle of reflection, how tall is the flagpole?

17 [Challenging] • Target: Review: Angles and angle measure • Cross-Lesson: 6.01 Polygons & Quadrilaterals

In ΔABC, line segment DE is drawn parallel to base BC such that it divides the triangle into two regions of equal area: $\text{Area}(\Delta ADE) = \text{Area}(BDEC)$. What is the ratio $\frac{AD}{AB}$?

18 [Challenging] • Target: Review: Angle relationships • Cross-Lesson: 7.01 Circles

In ΔABC, a transversal segment DE connects side AB at D and side AC at E such that $\angle ADE = \angle C$. If $AD = 3$, $AC = 9$, and the area of ΔADE is 8, what is the area of quadrilateral $BDEC$?

19 [Challenging] • Target: Points, lines, and planes • Cross-Lesson: 8.01 Area, Surface Area & Volume

In ΔABC shown below, point D is on AB and E on AC such that $\angle ADE \cong \angle C$. If $AD = 4$, $AE = 5$, and $AC = 12$, what is the length of side AB?

20 [Challenging] • Target: Line segments and distance • Cross-Lesson: 9.01 Transformations & Symmetry

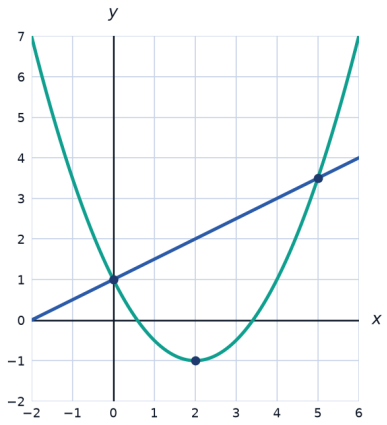
ΔABC and ΔDEF are shown below with $\angle A \cong \angle D = 55^\circ$. If $AB = 6$, $AC = 10$, $DE = 9$, and $DF = 15$, which similarity criterion applies and what is the scale factor from ΔABC to ΔDEF?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Points, lines, and planes

For the model $f(x) = 2x + 3$ in points, lines, and planes, evaluate $f(2)$.



a. 7

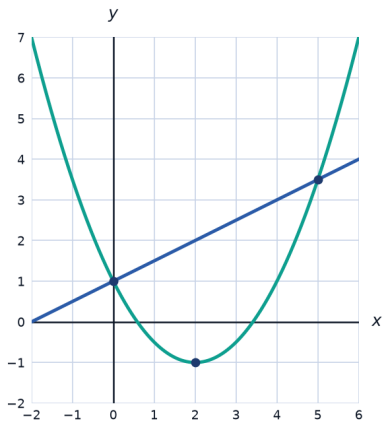
b. 9

c. 4

d. 11

22 [Foundational] • Target: Points, lines, and planes

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

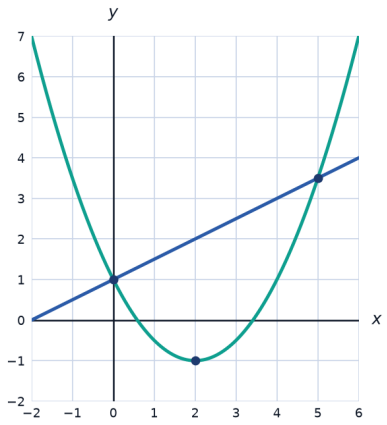
c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

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23 [Foundational] • Target: Points, lines, and planes

For the model $f(x) = 2x + 3$ in points, lines, and planes, evaluate $f(5)$.



a. 13

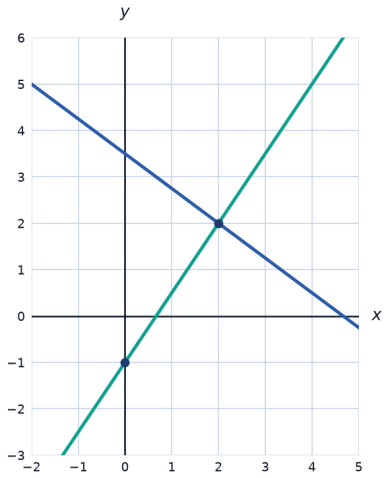
b. 15

c. 10

d. 17

24 [Foundational] • Target: Line segments and distance

For the model $f(x) = 3x + 4$ in line segments and distance, evaluate $f(2)$.



a. 10

b. 13

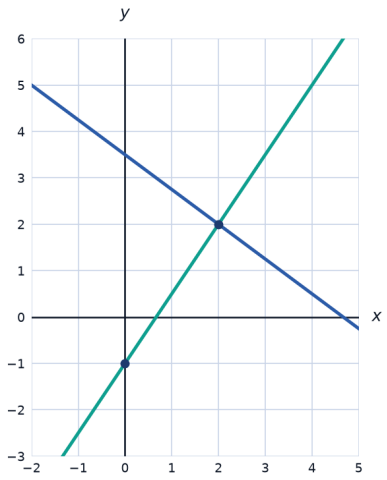
c. 6

d. 14

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25 [Foundational] • Target: Line segments and distance

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

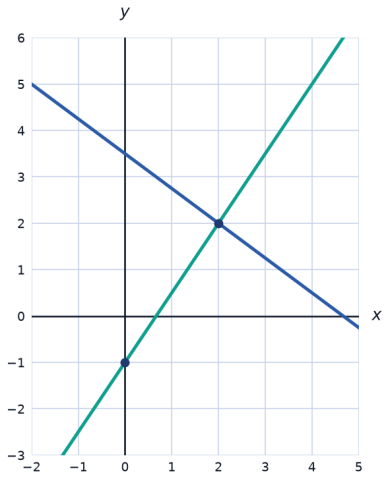
b. $x = 2$ and $x = 5$

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

26 [Foundational] • Target: Line segments and distance

For the model $f(x) = 3x + 4$ in line segments and distance, evaluate $f(5)$.



a. 19

b. 22

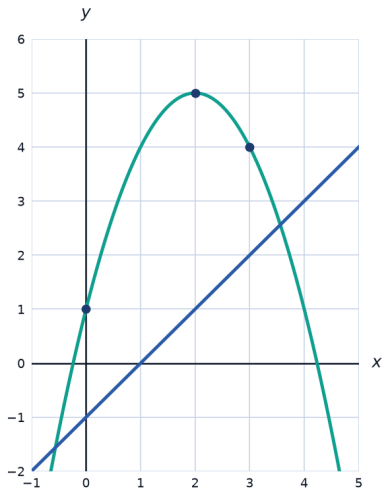
c. 15

d. 23

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27 [Foundational] • Target: Points and midpoints

For the model $f(x) = 4x + 5$ in points and midpoints, evaluate $f(2)$.



a. 13

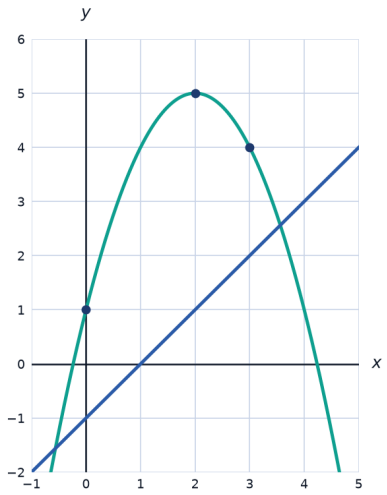
b. 17

c. 8

d. 17

28 [Foundational] • Target: Points and midpoints

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



a. $x = 2$ and $x = 7$

b. $x = 2$ and $x = 6$

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Points, lines, and planes

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

30 [Intermediate] • Target: Points, lines, and planes

For the model $f(x) = 2x + 3$ in points, lines, and planes, evaluate $f(3)$.

a. 9

b. 11

c. 6

d. 13

31 [Intermediate] • Target: Points, lines, and planes

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

32 [Intermediate] • Target: Line segments and distance

A data set recorded for line segments and distance consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

33 [Intermediate] • Target: Line segments and distance

For the model $f(x) = 3x + 4$ in line segments and distance, evaluate $f(4)$.

a. 16

b. 19

c. 12

d. 20

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34 [Intermediate] • Target: Line segments and distance

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ and $x = 7$

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

35 [Intermediate] • Target: Points and midpoints

For the model $f(x) = 4x + 5$ in points and midpoints, evaluate $f(3)$.

a. 17

b. 21

c. 12

d. 21

36 [Intermediate] • Target: Points and midpoints

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ and $x = 7$

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

37 [Intermediate] • Target: Review: Slopes of lines

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

a. $x = 1$ and $x = 7$

b. $x = 1$ and $x = 6$

c. $x = -1$ and $x = -6$

d. $x = 7$ and $x = 6$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Review: Slopes of lines • Cross-Lesson: 7.01 Circles

For the model $f(x) = 5x + 6$ in review: slopes of lines, evaluate $f(4)$.

a. 26

b. 31

c. 20

d. 30

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39 [Challenging] • Target: Review: Slopes of lines • Cross-Lesson: 8.01 Area, Surface Area & Volume

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

a. $x = 4$ and $x = 10$

b. $x = 4$ and $x = 9$

c. $x = -4$ and $x = -9$

d. $x = 13$ and $x = 36$

40 [Challenging] • Target: Review: Angles and angle measure • Cross-Lesson: 9.01 Transformations & Symmetry

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 7)$. What is its y-coordinate when $x = 1$?

a. 7

b. 11

c. 9

d. 11

41 [Challenging] • Target: Review: Angles and angle measure • Cross-Lesson: 10.01 Coordinate Geometry

In ΔABC studying review: angles and angle measure, $m\angle A = 34^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 87°

b. 97°

c. 79°

d. 146°

42 [Challenging] • Target: Review: Angles and angle measure • Cross-Lesson: 11.01 Probability

A geometric figure scaled by factor $k = 5$ has an original side length of 8 cm. What is the corresponding image length?

a. 35 cm

b. 40 cm

c. 48 cm

d. 200 cm

43 [Challenging] • Target: Review: Angle relationships • Cross-Lesson: 2.01 Lines & Angles

A geometric figure scaled by factor $k = 2$ has an original side length of 4 cm. What is the corresponding image length?

a. 6 cm

b. 8 cm

c. 12 cm

d. 16 cm

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44 [Challenging] • Target: Review: Angle relationships • Cross-Lesson: 3.01 Triangles & Congruence

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 8)$. What is its y-coordinate when $x = 2$?

a. 11

b. 16

c. 14

d. 17

45 [Challenging] • Target: Review: Angle relationships • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length?

a. 12 cm

b. 16 cm

c. 20 cm

d. 64 cm

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